

# OpenArm CAD file

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/home/egc365/Documents/cannabis-sample-prep-cell/torso/build\_openarm  
/openarm\_orca.step

577954123 bytes.

sha256

b2b5c751c709797d7bacc6f7bb0c128adab408a70c16278d4fa53e6af2fa7e0b.

The file is 578 MB. It is not uploaded. GitHub rejects a file over 100 MB.

The modular six-axis arm is not this CAD.

# The fork is egc365/openarm

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<https://github.com/egc365/openarm>

GitHub API on 2026-10-08: fork true. Parent enactic/openarm.

HTTP status 200.

Compare enactic:main...egc365:main: ahead 0, behind 10.

The 10 commits do not add a STEP tree.

The root is the website repo. It has no STEP tree and no STL tree.

enactic/openarm is the parent. It is not the owner's fork.

# That hardware URL is not the fork

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[https://github.com/egc365/openarm\\_hardware](https://github.com/egc365/openarm_hardware)

HTTP status 404. The saved response is in fork-status.txt.

A 404 does not say whether the name was never created, renamed, or deleted.

No fork of enactic/openarm\_hardware is owned by egc365. This URL is not the owner's fork.

The fork README names [https://github.com/enactic/openarm\\_hardware](https://github.com/enactic/openarm_hardware).

Release 2.0.0 was published 2026-09-09. The tar.gz is 163124751 bytes.

The local STEP was not compared byte for byte with that tar.gz.

# OpenArm motors

| Joint   | Motor  | Send | Recv |
|---------|--------|------|------|
| joint_1 | DM8009 | 0x01 | 0x11 |
| joint_2 | DM8009 | 0x02 | 0x12 |
| joint_3 | DM4340 | 0x03 | 0x13 |
| joint_4 | DM4340 | 0x04 | 0x14 |
| joint_5 | DM4310 | 0x05 | 0x15 |
| joint_6 | DM4310 | 0x06 | 0x16 |
| joint_7 | DM4310 | 0x07 | 0x17 |
| gripper | DM4310 | 0x08 | 0x18 |

CAN FD nominal 1 Mbps. CAN FD data 5 Mbps. Payload 6.0 kg peak and 4.1 kg nominal. Damiao list only. Anvil degrees are not in this table.

# Anvil range table

Degrees, read from the Anvil page image on 2026-10-08. Anvil is not the owner's fork.

| Joint   | Anvil 1.0    | Standard 2.0 | Anvil 2.0    |
|---------|--------------|--------------|--------------|
| J1      | -135 to +135 | -200 to +80  | -135 to +135 |
| J2      | -190 to +10  | -190 to +10  | -190 to +10  |
| J3      | -90 to +90   | -90 to +90   | -90 to +90   |
| J4      | 0 to 140     | 0 to 140     | 0 to 140     |
| J5      | -90 to +90   | -90 to +90   | -90 to +90   |
| J6      | -45 to +45   | -45 to +45   | -45 to +70   |
| J7      | -90 to +90   | -90 to +90   | -90 to +90   |
| Gripper | -45 to 0     | -45 to 0     | -45 to 0     |

# How to read the wrist table

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OpenArm 1.0 uses J6 for extension and flexion, and J7 for radial and ulnar deviation.

OpenArm 2.0 swaps those jobs. J6 is radial and ulnar deviation. J7 is extension and flexion.

Flexion therefore moves from J6 at -45 to +45 onto J7 at -90 to +90.

The J6 row alone does not show a larger flexion number. Standard 2.0 J6 stays -45 to +45.

Anvil 2.0 J6 is -45 to +70. It is not -70 to +70. The callout says Extra range in J6.

Standard 2.0 J1 is -200 to +80. Both Anvil columns keep J1 at -135 to +135.

These degrees are the Anvil page. They are not a measured stop on the local STEP.

# Anvil page, joint limits

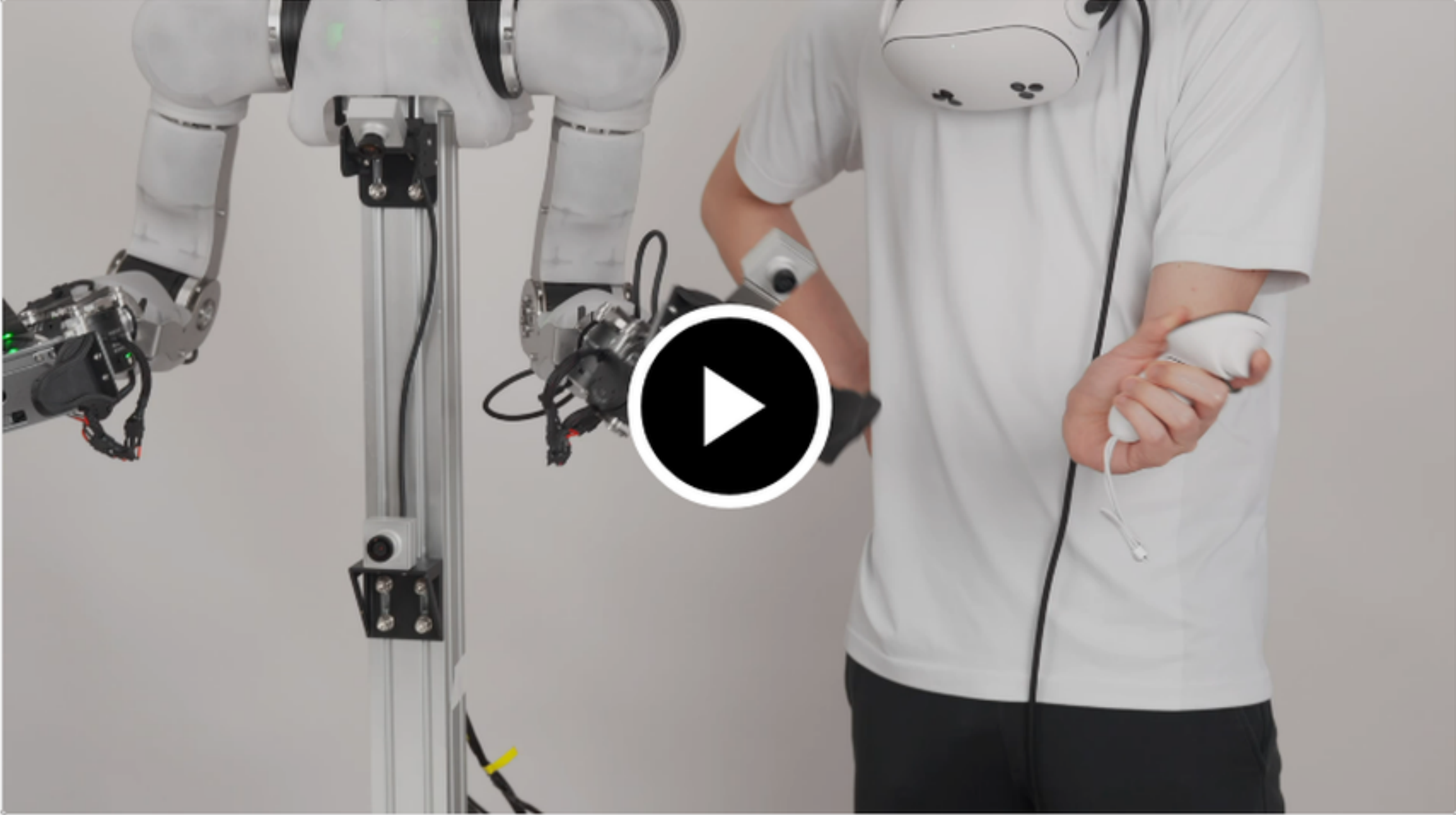
OpenARM 1.0 wrist motion hits joint limits. The arm and the elbow make large corrective moves.

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and far more capable on real bimanual tasks.

## Smoother, more human-like motion

With **OpenARM 1.0**, normal teleoperation wrist motions often hit the joint limits due to a mismatch between the human kinematic and the robot kinematic. To compensate, the arm and elbow had to make large corrective movements to achieve certain poses, resulting in jerky, unnatural motion.



The new **Anvil OpenARM 2.0** moves through the same wrist motions smoothly, with no compensating motion needed in the arm or elbow. This is because the joints now support the full range of human wrist motions, meaning bimanual movements stay fluid even when both arms are working together near the body, expanding the OpenARM's viable workspace.

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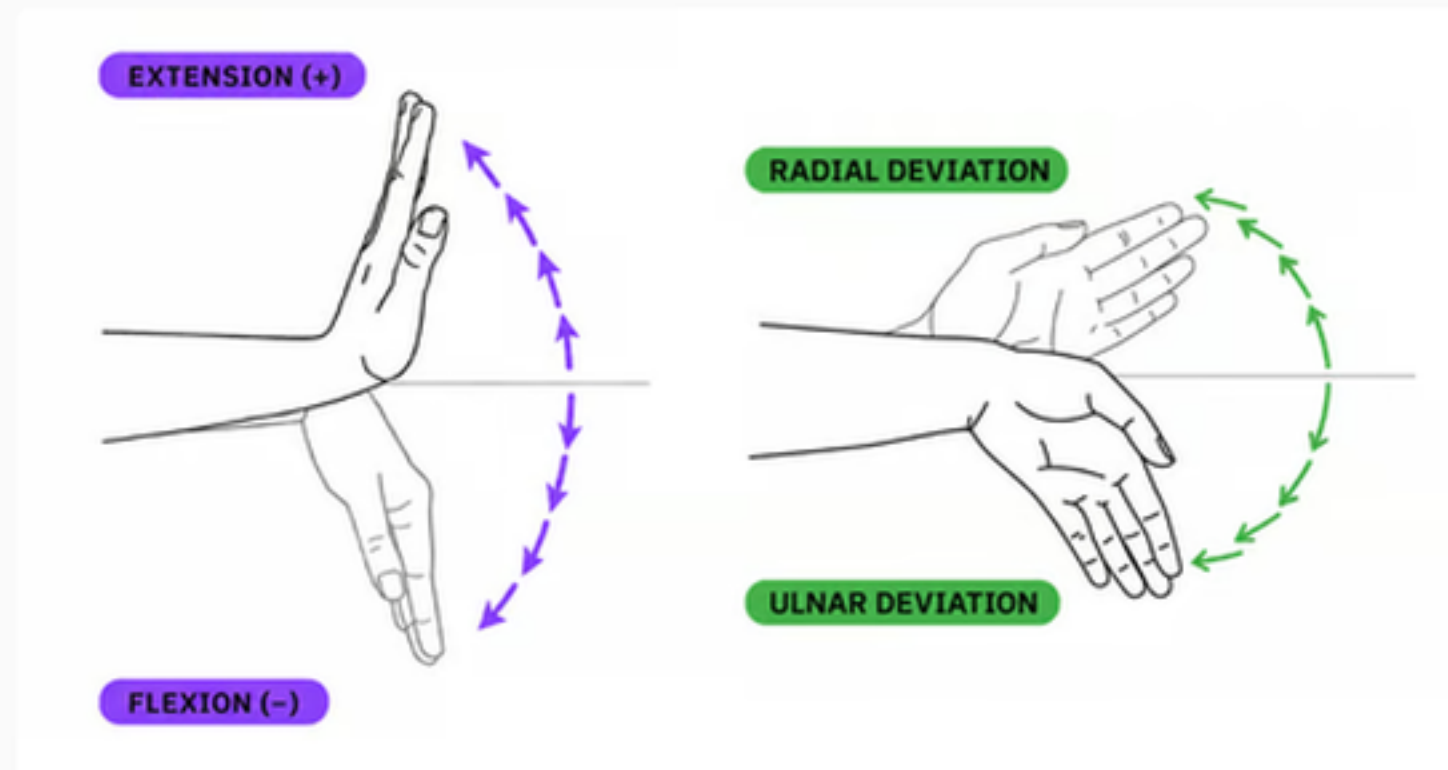
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# Anvil page, human wrist

Flexion and extension are the larger human motion. OpenArm 2.0 swaps J6 and J7 so flexion gets that range.

> extension, and the other that controls radial/ulnar deviation. As you can see in the diagram below, humans naturally have a much larger range of motion in flexion/extension.



improved wiring

A small change where it counts

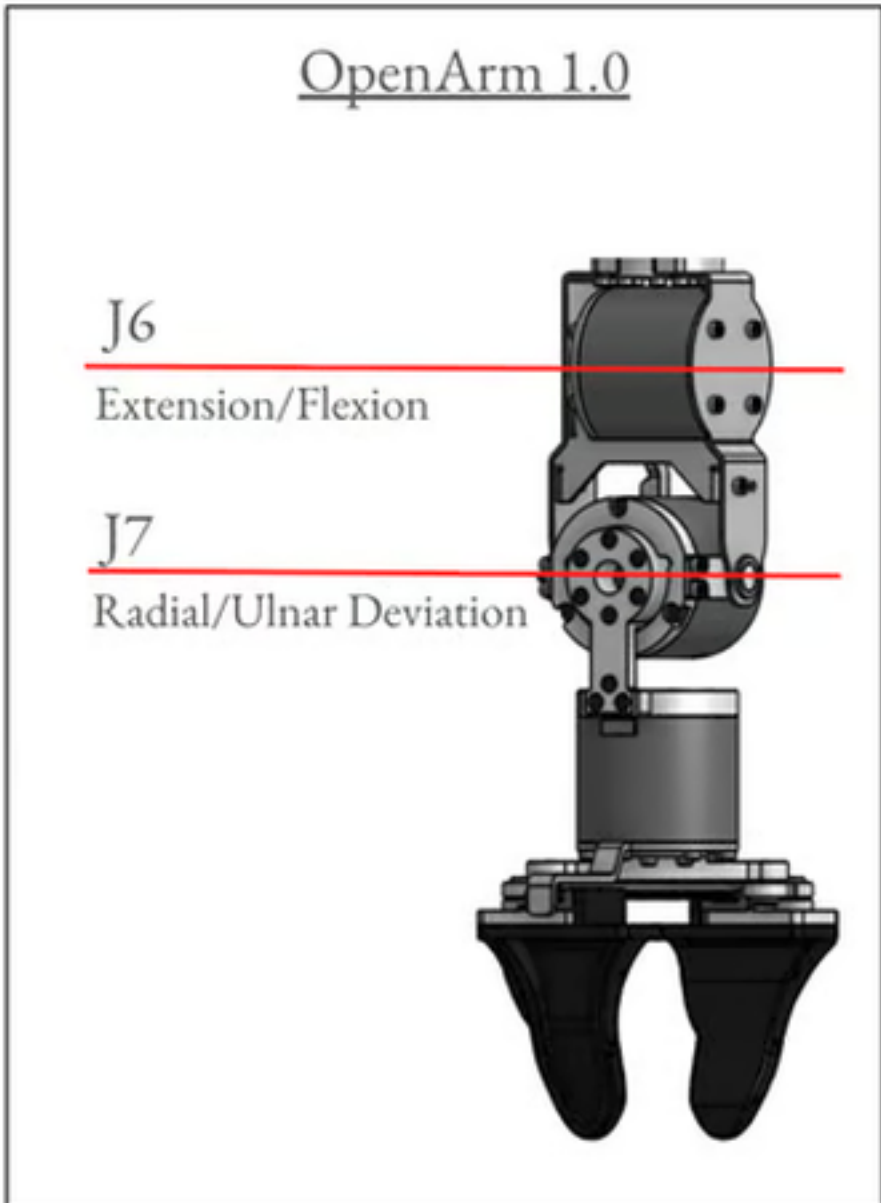
> In the original OpenArm 1.0, J6 controlled the wrist's extension/flexion and J7 (the joint with a wider range of motion) controlled the wrist's radial/ulnar deviation. For OpenArm 2.0, we swapped these joints to allow for a much larger range of motion for extension and flexion, mimicking human range of motion. We also updated our new J6 joint to support a slightly wider

# Anvil page, joint drawing

OpenArm 2.0 puts radial and ulnar deviation on J6 and extension and flexion on J7. Anvil says Extra range in J6.

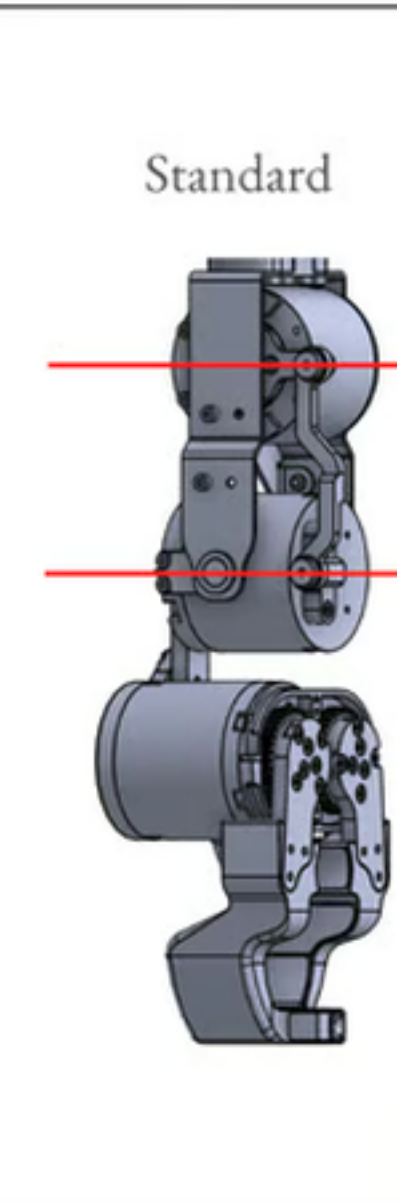
mimicking human range of motion. we also updated our new 30 joint to support a slightly wider radial/ulnar deviation range.

OpenArm 1.0

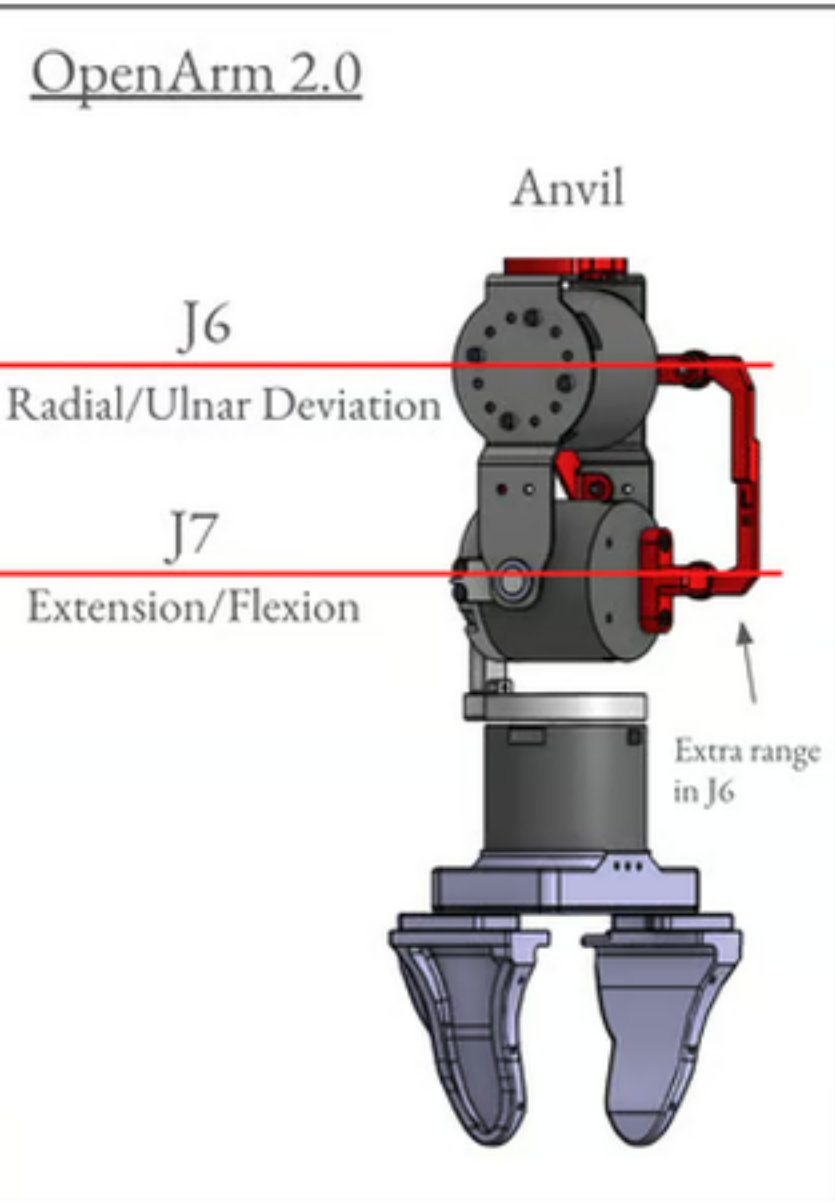


OpenArm 2.0

Standard



Anvil



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|         | Anvil OpenARM 1.0 | Standard OpenARM 2.0 | Anvil OpenARM 2.0 |
|---------|-------------------|----------------------|-------------------|
| J1      | -135° to + 135°   | -200° to +80°        | -135° to + 135°   |
| J2      | -190° to +10°     | -190° to +10°        | -190° to +10°     |
| J3      | -90° to +90°      | -90° to +90°         | -90° to +90°      |
| J4      | 0° to 140°        | 0° to 140°           | 0° to 140°        |
| J5      | -90° to +90°      | -90° to +90°         | -90° to +90°      |
| J6      | -45°to +45°       | -45° to +45°         | -45° to +70°      |
| J7      | -90° to +90°      | -90° to +90°         | -90° to +90°      |
| Gripper | -45° to 0°        | -45° to 0°           | -45° to 0°        |

EXTENSION (+)



FLEXION (-)

RADIAL DEVIATION

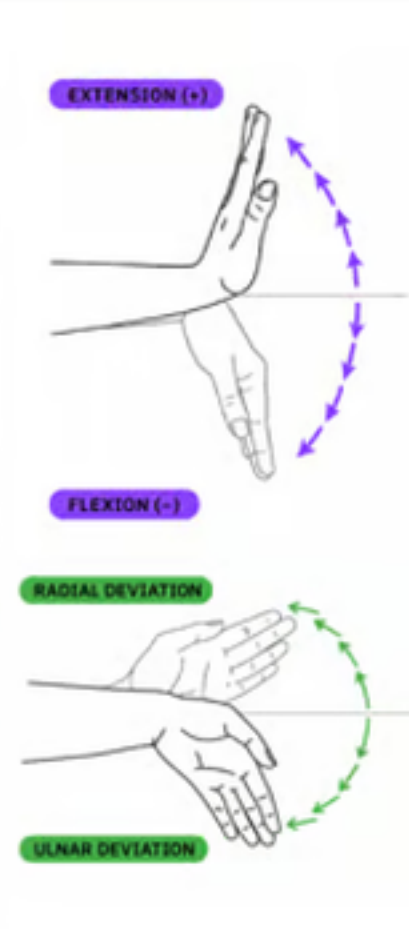


ULNAR DEVIATION

## Anvil page, table photo

The red boxes are the two J6 cells for Standard OpenARM 2.0 and Anvil OpenARM 2.0.

|         | Anvil OpenARM 1.0 | Standard OpenARM 2.0 | Anvil OpenARM 2.0 |
|---------|-------------------|----------------------|-------------------|
| J1      | -135° to + 135°   | -200° to +80°        | -135° to + 135°   |
| J2      | -190° to +10°     | -190° to +10°        | -190° to +10°     |
| J3      | -90° to +90°      | -90° to +90°         | -90° to +90°      |
| J4      | 0° to 140°        | 0° to 140°           | 0° to 140°        |
| J5      | -90° to +90°      | -90° to +90°         | -90° to +90°      |
| J6      | -45° to +45°      | -45° to +45°         | -45° to +70°      |
| J7      | -90° to +90°      | -90° to +90°         | -90° to +90°      |
| Gripper | -45° to 0°        | -45° to 0°           | -45° to 0°        |



## Harder tasks made easy

The extended range does more than smooth out wrist motion. When combined with a well tuned IK solver, it opens up workspace that used to be awkward or out of reach, especially close to the robot's body, where so much real manipulation happens. Tasks that were tedious or nearly impossible on OpenARM 1.0 are now well within reach: fastening zip ties, disassembling Legos, and other fine, two-handed work that depends on getting the wrist into the right orientation near the body.



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# Anvil page, named tasks

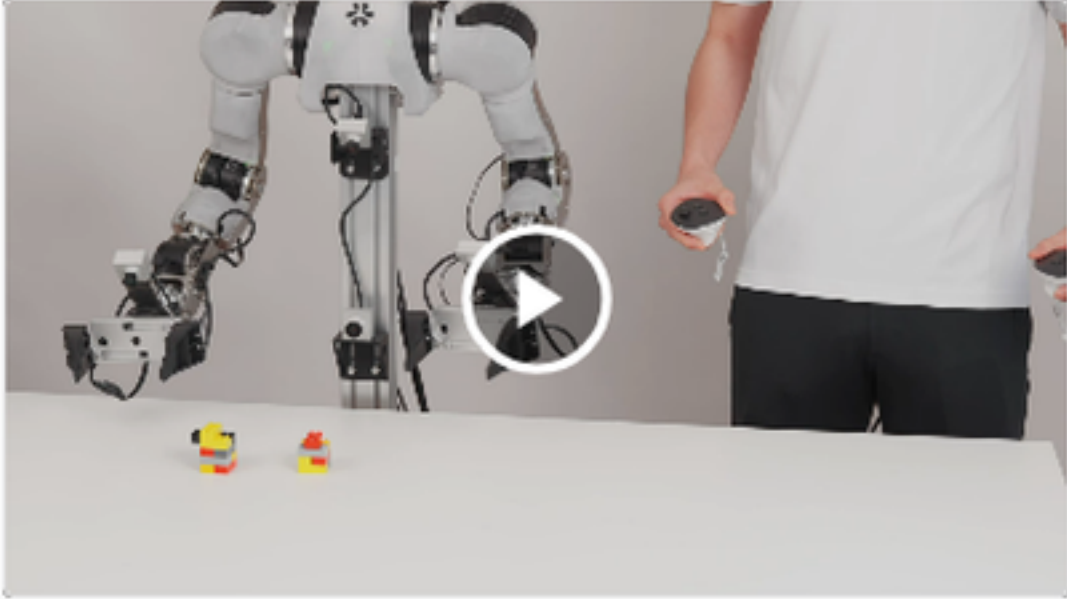
Fasten 4 zip ties in 1 min. Disassemble complex Legos. Align and insert cables. Pack raw eggs.

### Harder tasks made easy

The extended range does more than smooth out wrist motion. When combined with a well tuned IK solver, it opens up workspace that used to be awkward or out of reach, especially close to the robot's body, where so much real manipulation happens. Tasks that were tedious or nearly impossible on OpenARM 1.0 are now well within reach: fastening zip ties, disassembling Legos, and other fine, two-handed work that depends on getting the wrist into the right orientation near the body.



Fasten 4 Zipties in 1 min



Disassemble complex legos



Easily align and insert cables



Pack raw eggs

### Improved Wiring

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# Anvil page, cable still

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One video frame from the cable task. This frame does not add a joint number.

>



>

improved wiring

A small change where it counts

# Anvil page, three cables

Original OpenARM, Anvil OpenARM 1.0, Anvil OpenARM 2.0. The footer says Last updated 2 months ago.

arm that holds up to the demands of continuous data collection. This solves any prior issues with motors dropping from the control bus.



Original  
OpenARM

Anvil  
OpenARM 1.0

Anvil  
OpenARM 2.0

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## A small change where it counts

It's a small set of mechanical and cable changes with an outsized impact on teleoperation. The more naturally an operator can move, the less they have to fight the hardware, and the higher-quality the demonstration data they collect for training Physical AI models.

Anvil OpenARM 2.0 keeps everything that makes OpenARM easy to build, modify, and repair, now with a wrist that moves the way you do. For full specs and assembly details, check out our [docs](#).

Last updated 2 months ago

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